ECM3175 - 3rd Year Individual Projects

Final Report Writing Guidelines

09.03.2026

Module leader: Dr Evangelos Papatheou

Email: e.papatheou@exeter.ac.uk

Session agenda (2-hours)

- **Lecture:** Introduction to Final Report Writing
- **Questions & Answers** session
- **Break**
- **Workshop:** How to write your project (adapted from Dr Rachel Canter-study zone)

Final report

- **Final report (90% final mark)**
  - 35-page single pdf file (size limit 15MB)
  - Deadline 1 May 2026
  - ELE (includes automatic Turnitin submission):
- **Interview (10% final mark)**
  - will be organised by your supervisor
  - 21-29 May 2026
  - Please make sure you are available for the entire week

Final Report Format

1. Abstract
2. Introduction and background
3. Literature review (should not merely be duplication of preliminary version, but should be updated and expanded)
4. Methodology and theory
5. Experimental work/analytical investigation/design
6. Experimental or analytical results/description of final constructed product
7. Discussion and conclusions
8. Project management, consideration of relevant sustainability and health and safety issues where appropriate

Final Report Marking

- Check marking sheet
- Report template
  - Can be found on ELE

Final Report – Project Assessment Criteria- 3rd Year Individual Projects (ECM3175 and ECM3149)

Section	Excellent (>80%)	Very Good (70-80%)	Good (55-69%)	Satisfactory (40-54%)	Poor (below 40%)
Abstract (5 Marks) • Background to project • Project aim • Methodologies used • Main outcomes • Conclusions	• Very well written, accurately and concisely captures all the essential aspects of the project, methodology, outcomes and issues	• Reasonably well written and captures most of the essential elements of the project, methodology, outcomes and issues	• Adequately written and captures most elements methodology, outcomes and issues though missing some information	• Poorly written and does not clearly convey information concerning project topic, method, issues and/or outcomes	• Badly written and/or does not summarise the project topic and its outcomes
Introduction and background (5 Marks) • What • Why • How • Aim • Objectives	• Thorough and in-depth understanding of the background • Clear, logical and crisp overview of the problem being addressed • Aim and objectives are precise, outlining clearly the what, why and how without leaving room for ambiguities	• Good, consistent knowledge and understanding of the background • Logical and unambiguous overview of the problem being addressed • Clear, well-defined aim and objectives	• Sound, routine knowledge of the problem being addressed • An appropriate overview of the problem being addressed but with minor shortcomings • Aim and objectives are stated but with minor ambiguities	• Broadly accurate understanding of the problem being addressed • Some elements missing and flaws evident • Aim and objectives are given but need much sharpening	• Fails to demonstrate a basic understanding of the problem • Reader left confused about the focus of project. • Lacks/badly written aim and objectives
Literature review (15 Marks) • What has been done before • How it shapes your project • Good referencing	• Exceptionally wide range of relevant literature used critically to inform argument, balance discussion and/or inform problem-solving. • Consistently accurate use of references. • Shows follow up of most recent advances	• Critical engagement with appropriate reading. • Knowledge of research-informed literature. • Consistently accurate use of references.	• Knowledge of literature beyond core text(s). • Literature used accurately but descriptively. • Referencing generally sound.	• Some evidence of reading, with superficial linking to given text(s). • Referencing is largely consistent, but with some weaknesses.	• Evidence of little reading and/or indiscriminate use of sources. • Very few references and/or badly cited.
Methodology and theory (20 Marks) Elaboration of project-relevant key concepts identified in literature review	• Excellent description of the methodology and/or experimental/design procedure • Demonstrates a thorough and in-depth knowledge and understanding of concepts and theories that goes beyond expectations for this level • Extensive, relevant and logically organised • Shows a good degree of originality/creativity with notable progress beyond background reading • Clear evidence of significant progress towards objectives	• Very good description of the methodology and/or experimental/design procedure • Good consistent knowledge and understanding of concepts and theories • Relevant and logically organised • Shows some degree of originality with clear and useful progress beyond background reading • Clear evidence of good progress towards stated objectives	• Good description of the methodology and/or experimental/design procedure with minor flaws • Sound understanding of concepts and theories • Acceptable coverage • Shows reasonable progress beyond background reading • Evidence of an acceptable level of progress towards stated objectives	• Appropriate description of the methodology and/or experimental/design procedure with evident of flaws • Broadly accurate knowledge and understanding of the concepts and theories • Limited coverage • Shows limited progress beyond background reading • Some evidence of progress towards stated objectives	• Incomplete/incorrect description of the methodology and experimental/design procedure • Lack of understanding of concepts and theories • Shows no progress beyond background reading • No evidence of progress towards objectives

Final Report – Project Assessment Criteria- 3rd Year Individual Projects (ECM3175 and ECM3149)

Experimental work /analytical investigation/design (20 Marks) • Analysis, experiments or design (hardware, software, procedures) • Design (of equipment or procedure)	• Excellent description of how the methodology/experimental work is employed/conducted • Can collect and interpret appropriate data/ information and undertake research tasks with autonomy and exceptional success.	• Good description of how the methodology/experimental work is employed/conducted • Can collect and interpret appropriate data and successfully undertake research tasks with a degree of autonomy.	• An acceptable description of the methodology/experimental work is employed/conducted • Can collect and interpret appropriate data/ information and undertake straightforward research tasks with external guidance.	• Limited description of the methodology/experimental work is employed/conducted • Some evidence of ability to collect appropriate data/information and undertake straightforward research tasks with external guidance.	• Poor presentation of the methodology/experimental work is employed/conducted • Limited evidence of skills in the range identified for the assessment at this level. • Significant weaknesses evident.
Presentation of experimental or analytical results/description of final constructed product (15 Marks) Exposition of outcomes of the project in the form of a finished product, set of experimental results or outputs from computational or theoretical work	• Neat, logical and crisp presentation of results as evidence of thorough and detailed work • Excellent plots, tables, drawings presented creatively with clear labelling and clear references from text • Outcomes self-evident from the presentation without having to explore through text • Results show work beyond expectations at this level.	• Clear, unambiguous presentation of results as evidence of a good piece of technical work • Neat plots, tables, drawings with clear labelling and clear references from text • Outcomes self-evident from the presentation without having to explore through text	• Technically correct presentation of results but with minor shortcomings • Good legible plots, tables, drawings that are neatly linked from the text but with minor flaws • Major outcomes are largely evident from the presentation	• Results are given but with some evident gaps • Purpose of some plots, tables, drawings etc. is not clear • Presentation is not coherent • Some conclusions are brought out	• Few results presented • Fundamental technical issues present • Drawings, tables, plots hard to read • Inadequate information to substantiate conclusions
Discussion and conclusions (10 Marks) • Description of what has been learnt from experimental/analytical work or design • Distillation of important findings and their significance • Specific observations to inform future research	• Excellent discussion that distils new knowledge from results • A concise set of statements of convincing conclusions • Clearly defines foundations made for future research or product development • Conclusions demonstrate novel and publishable quality work	• Technically sound discussion highlighting findings from this study with well-balanced arguments. • Arguments generally logical, coherently expressed, well organised and supported. • Concise set of statements outlining sound conclusions. • Clear set of observations to inform future research.	• Good discussion highlighting the important findings but with minor weaknesses. • An emerging awareness of different stances and ability to use evidence to support a coherent argument. • Broadly valid conclusions.	• Sense of argument emerging though not completely coherent. • Some evidence to support views, but not always consistent. • Some relevant conclusions.	• For the most part descriptive with no clarity. • Important results not highlighted and views/findings sometimes illogical or contradictory • Generalisations/statements made with limited evidence. • Conclusions lack relevance and/or validity.
Project management, consideration of relevant sustainability and health and safety issues where appropriate, overall presentation throughout report (10 Marks) • Procedures and systems for project management • Sustainability/H&S as appropriate	• Very well defined how the project was managed. • Shows excellent awareness and relevance of health and safety issues, • Demonstrates a thorough understanding of sustainability issues and their relevance, • Excellent level of English	• Clearly defined how the project was managed, • Shows awareness and relevance of health and safety issues, • Demonstrates full understanding of sustainability issues and their relevance, • Few to no mistakes • High standard of English	• Sound overall management of project with minor shortcomings • Sufficient coverage of health and safety and sustainability issues but few elements missing • Good level of English with few minor spelling and grammatical errors	• Acceptable overall level of management however with evident flaws • Some coverage of health and safety and sustainability issues with some elements missing • Readable but contains some typos and grammatical errors	• Little clarity on what the student did • Unconvincing management strategy • Does not cover relevant health and safety and sustainability issues • Difficult to follow with lots of typos and grammatical mistakes

Thesis Presentation

- Report structure
- Report formatting
- Use of tables, figures and equations
- Spelling

Figures and Tables

- Each figure and table should be **numbered** respectively
- Each figure or table should have a **caption**
- Each figure or table should be **referred in text**
- The key points showed in a figure/table should be **explained in the text**
- <u>Cite source!! (if applicable)</u>

Figures - Example

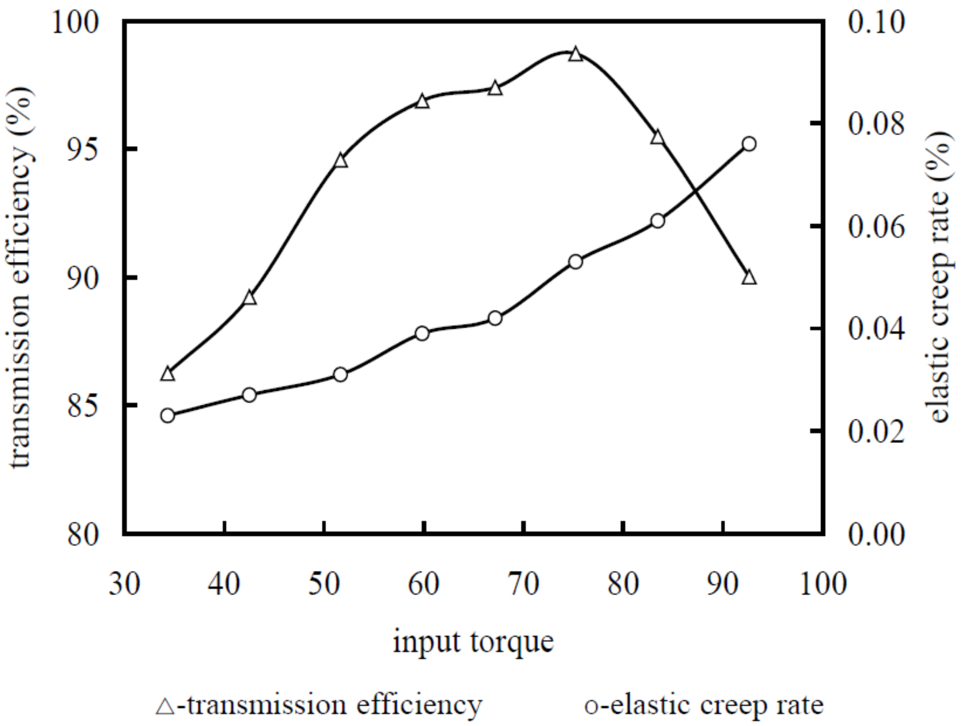

Fig. 10 Experimental results

Tables - Example

Table 2 Contact ratio of the cosine and the involute gear drive

tooth number Z_1	tooth number Z_2	modulus m	cosine gear drive	involute gear drive
15	32	3 mm	1.2642	1.5745
17	40	3 mm	1.2826	1.6142
21	60	3 mm	1.3402	1.6769

References

- Any work, ideas, data, or images other than yours must be **acknowledged**
- Failing to do this will be regarded as **plagiarism**
- Reference Systems
  - Numerical Citation System
  - Harvard Citation System

Numerical Citation System

- In the main body of text, use a sequence number in **square brackets** to refer a source in the order it first appears.

  …[1]…[2]…[3,4]…[1-3,5]…

- In the 'References' section, the articles are listed in the **same sequence order**.

References

[1] Wright, C.D., 2002, …

[2] Wu, Y., G. Frizelle, and J. Efstathiou, 2007, …

[3] Smith, C. and K. Evans, 2005, …

Numerical System - Example

Some authors [4,8–11] suggest that agent-based systems may be enriched, obviously at the expense of higher implementation complexity, by defining special roles for facilitating system operation. Such

References

[1] G. Dosi, Technological paradigms and technological trajectories, Res. Pol. 11 (1982) 147–162.

[2] R. Davis, R.G. Smith, Negotiation as a metaphor for distributed problem solving, Artificial Intel. 20 (1983) 63–109.

Harvard Citation System

- In the main body of text, use **bracketed surname** of the authors and the year of publication.
  - …(Wright 2002, Smith and Evans 2005, Wu *et al*. 2007)
  - Wright (2002) …
- In the references, the articles are listed in **alphabetical order** and then in year order.

  - **References**
- Smith, C. and K. Evans, 2005, …
- Wright, C.D., 2002, …
- Wright, C.D., 2004, …
- Wu, Y., G. Frizelle, and J. Efstathiou, 2007, …

Single author

Two authors

More than two

Background

Most of the currently available WDN hydraulic simulators, such as *EPANET* (Rossman 2000a), work as DDA-type solvers utilizing the GGA (Todini and Pilati 1988). However, these simulators can be used to analyze WDNs under pressure-deficient conditions as well, which can be done in two principal ways.

The first approach relies on using a specific pressure-demand relationship that requires changing the source code of the simulator. A number of examples of this approach exist. Ackley et al. (2001) and Yoo et al. (2012) presented optimization methods that maximized nodal outflow under abnormal operating conditions. Although they delivered results for simple WDNs, no generic solution approach was developed for practical applications. More recently, Goldstein's algorithm was used by Elhay et al. (2015) to upgrade different NHFRs in the GGA (Todini and Pilati 1988). This approach was successfully demonstrated on eight challenging (i.e., one has 20,000 pipes) case study networks. However, this involved only steady-state hydraulic analysis. Rossman (2000b) extended *EPANET* by implementing a flow emitter at a node to simulate demand delivered as a function of available pressure. However, a flow emitter produces negative outflow (i.e., inflow) when nodal pressure becomes negative and there is no upper limit for the discharge value. Cheung et al. (2005) modified the source code of *EPANET* by introducing emitters into the network model to simulate pressure-driven demands by using an object-oriented toolkit (OOTEN). Yet this approach failed to converge when attempting to model highly looped WDNs under low flow conditions and EPS analysis. A modification of the emitter methodology was proposed by Morley and Tricarico (2014) through *EPANETpdd* by allowing each emitter to have its own empirical exponent. However, *EPANETpdd* fails due to convergence-related issues when applied to medium and larger or more complex WDNs. Liu et al. (2011), Siew and Tanyimboh (2012), and Jun and Guoping (2013)

Final report submission

- 35-page single pdf file (size limit 15 MB)
- Deadline 1 May 2026
- ELE (Turnitin)

- All questions concerning ELE, document submission, Turnitin etc. should be addressed to Harrison Student Services